NGUYỄN GIA KHÔI

Backend Engineer

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PROFESSIONAL SUMMARY

Performance-driven Backend Engineer specializing in AI-powered architectures, distributed data pipelines, and GraphRAG systems. Proven track record in optimizing production database latency from ~15s to sub-second execution, implementing distributed concurrency control with Redis, and building domain-grounded LLM retrieval workflows using FastAPI, Neo4j, and LangGraph. Co-author of 2 published scientific papers in applied AI.

WORK EXPERIENCE

Backend Engineer Intern **09/2025 - 01/2026**

Phuong Quan Trading Service

- Consolidated fragmented data architectures from 3 legacy internal systems into a unified backend service, slashing end-to-end API response latency from 10–15 seconds to sub-second execution.
- Optimized complex PostgreSQL queries and index structures, restructuring execution plans to reduce query runtime from ~15 seconds to under 300ms for high-volume transactions.
- Architected a distributed locking mechanism using Redis to eliminate race conditions during concurrent multi-source synchronization of 1M+ records, guaranteeing strict transactional data consistency.
- Use Grafana to monitoring dashboards and centralized log analysis to track production throughput, detect bottlenecks, and accelerate root-cause debugging for critical backend workflows.

Document Management System – Nam Saigon Polytechnic College

Backend Developer | Node.js, Express, MongoDB, Google Drive API

- Engineered a role-based internal document workflow platform, implementing granular RBAC permissions for multi-department approval cycles and access auditing
- Integrated Google Drive API as an automated cloud object storage layer, reducing infrastructure overhead and storage costs while ensuring secure file indexing and lifecycle management.
- Designed normalized MongoDB schemas and RESTful endpoints to handle high-frequency document metadata tracking, status transitions, and fast audit trail retrieval.
- Analyzed requirement changes by evaluating impact on system design and data integrity; proactively resolved ambiguities by working with stakeholders to define data flows and essential fields

PROJECT

TasttyMind – Graph RAG-powered Nutrition Chatbot

Team Lead | FastAPI, Neo4j, LangGraph, Express.js

- Architected an end-to-end GraphRAG pipeline combining structured clinical knowledge with LLM-based reasoning to generate personalized dietary plans for patients with non-communicable chronic diseases.
- Designed and populated a hierarchical Neo4j Knowledge Graph (~650 nodes, 1,700 relationships) modeling complex interactions between medical contraindications, nutrients, and food sources.
- Developed a hybrid retrieval-generation engine utilizing Cypher graph traversal, LLM re-ranking, and LangGraph state orchestration to supply grounded, context-aware prompts.
- Engineered defensive prompt constraints and validation guardrails, mitigating hallucination rates and enforcing strict fallback handling for out-of-scope clinical queries.
- Built high-throughput asynchronous backend microservices using FastAPI to handle concurrent vector/graph queries and stream low-latency LLM responses.

Tour Booking Web Application

Fullstack / Backend Lead | Node.js, Express, Redis, MongoDB

- Engineered a high-concurrency reservation engine with a 15-minute seat lock using Redis distributed locks, completely preventing double-booking and race conditions during peak traffic.
- Designed schema architecture in MongoDB supporting multi-tier pricing policies (adult/child), dynamic date-based inventory, and automated e-ticket issuance.
- Implemented Role-Based Access Control (RBAC) across multi-tenant user flows (End Users, Tour Operators, Platform Administrators) with secure authentication and mock payment webhooks.
- Developed optimized RESTful search and filtering APIs supporting multi-parameter queries (destination, date, price range) with cached search results to minimize database load.

https://github.com/nguyen-gia-khoi/GlobeTrek

EDUCATION

Publications

- 1
- 2

Bachelor of Information Technology

- University of University of Foreign Languages - Information Technology
- Final GPA: 3.25
- IELTS 6.5

2022 - 2026

TECHNICAL SKILLS

- Languages: Python, JavaScript/Node.js, SQL, Cypher (Neo4j)
- Backend: FastAPI, Express.js, RESTful API Design, Asynchronous Programming
- Databases & Caching: PostgreSQL, MongoDB, Redis (Distributed Locking, Caching), Neo4j (Graph DB)
- AI & LLM Systems: GraphRAG, LangGraph, RAG, OpenAI API
- DevOps & Infrastructure: Docker, Grafana (Monitoring & Metrics), Git, Postman